Python Programming - 2301CS404

Lab - 10

Roll No : 459

Name : Vibhu Shihora

01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string

- line by line

- in the form of a list

In [10]:
```
fp = open("f1.txt" , "r")

# print(fp.read)  # Its Read Whole File

n = fp.readline()
while(n != ''):
    print(n,end="")
    n = fp.readline()

# print(fp.readlines())  #All The Lines As List

fp.close()
```

```
Hello Jay
Ok
By
```

02) WAP to create file named "new.txt" only if it doesn't exist.

In [12]:
```
fp = open("f2.txt","w")
```

```
fp.close()
```

03) WAP to read first 5 lines from the text file.

In [19]:
```
fp = open("f1.txt","r")

count = 0

while(count != 5):
    print(fp.readline() , end="")
    count += 1

fp.close()
```

```
Hello Jay
Ok
By
line 4
line 5
```

04) WAP to find the longest word(s) in a file.

In [21]:
```
fp = open("f1.txt","r")

words = fp.read().split(" ")

longestWord = ""

for w in words:
    if(len(longestWord) < len(w)):
        longestWord = w

print(longestWord)

fp.close()
```

```
DarshanUniversity
```

05) WAP to count the no. of lines, words and characters in a given text file.

In [30]:
```
fp = open("f1.txt","r")

n = fp.readline()

count = 0

while(n != ''):
    count += 1
    n = fp.readline()

fp.seek(0,0)

totalWords = len(fp.read().split(" "))

fp.seek(0,0)
```

```
totalChar = len(list(fp.read()))

print("Total Numbers Of Lines : " , count)
print("Total Numbers Of Words : " , totalWords)
print("Total Numbers Of Chars : " , totalChar)

fp.close()
```

```
Total Numbers Of Lines :  8
Total Numbers Of Words :  6
Total Numbers Of Chars :  71
```

06) WAP to copy the content of a file to the another file.

In [35]:
```
fp = open("f1.txt","r")

fp2 = open("f2.txt","w+")

content = fp.read()

fp2.write(content)
fp2.seek(0)
print(fp2.read())

fp.close()

fp2.close()
```

```
Hello Jay DarshanUniversity
Ok
By
line4
line5
ok2 (line6)
line7
line8
```

07) WAP to find the size of the text file.

In [36]:
```
fp = open("f1.txt","r")

print(len(fp.read()))
```

```
71
```

08) WAP to create an UDF named frequency to count occurances of the specific word in a given text file.

In [47]:
```
def frequency(words):
    return words.count("hey")

fp = open("f2.txt","r")

print(frequency(fp.read().split()))

# fp.seek(0)
# print(fp.read().split())
```

```
fp.close()
```

```
3
['Hello', 'Jay', 'DarshanUniversity', 'Ok', 'hey', 'By', 'line4', 'line5', 'ok2',
'(line6', 'hey', 'line7', 'line8', 'hey']
```

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

In [21]:
```
fp = open("f2.txt","w+")

marks = input("Enter Subject Marks").split(" ")

# print(marks)

for i in range(0,len(marks)):
    fp.write("Marks Of Subject"+str(i)+":"+marks[i]+"\n")

fp.seek(0)

li = []

for i in fp:
    print(int(i.split(":")[-1][0:2]))
    li.append(int(i.split(":")[-1][0:2]))


print("max : ",max(li))

# or

# n = fp.readline().split(":")[-1]

# while(n != ''):
#     print(n,end="")
#     n = (fp.readline().split(":"))[-1]

fp.close()
```

```
77
88
55
99
99
```

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

In [1]:
```
def isPrime(n):

    if(n < 2):
        return False

    for i in range(2 , int(pow(n,0.5)) +1):
```

```
        if(n % i == 0):
            return False
    else:
        return True


with open("primenumbers.txt","w") as fp:
    count = 0
    n = 2

    fp.write("No\tPrimeNo\n")

    while(count != 100):
        if(isPrime(n)):
            fp.write(str(count+1) + "\t"+str(n) + "\n")
            count += 1
        n += 1
```

11) WAP to merge two files and write it in a new file.

In [4]:
```
fp = open("f1.txt","r")
fp2 = open("f2.txt","r")
fp3 = open("f3.txt","w")

fp3.write(fp.read() + "\n" + fp2.read())

fp.close()
fp2.close()
fp3.close()
```

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

In [7]:
```
fp = open("f1.txt" , "r")
fp2 = open("f4.txt" , "w")

fp2.write(fp.read().replace("hello","heyy"))

fp.close()
fp2.close()
```

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

In [16]:
```
with open("demo.txt", "w") as f:
    f.write("ABCDEFGHIJKLMNOPQRSTUVWXYZ")

# binary mode mate b
fp = open("demo.txt", "rb")

# 1. From Beginning (seek(offset,0))
print("Initial position:", fp.tell())

fp.seek(5, 0)      # move 5 bytes from beginning
```

```python
print("After seek(5,0):", fp.tell())

print("Reading next char:", fp.read(1))   # Should read F


# 2. From Current Position (seek(offset,1))
fp.seek(3, 1)      # move 3 bytes forward from current
print("\nAfter seek(3,1):", fp.tell())

print("Reading next char:", fp.read(1))   # Should read J


# 3. From End (seek(offset,2))
fp.seek(-4, 2)     # move 4 bytes backward from end
print("\nAfter seek(-4,2):", fp.tell())

print("Reading next char:", fp.read(1))   # Should read W

fp.close()
```

```
Initial position: 0
After seek(5,0): 5
Reading next char: b'F'

After seek(3,1): 9
Reading next char: b'J'

After seek(-4,2): 22
Reading next char: b'W'
```